

# The Igusa Square Root

## The Companion

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July 2026

### Abstract

*The Igusa cusp form  $\chi_{10}$  of weight ten on  $\mathrm{Sp}_4(\mathbb{Z})$  has a square root: the weight-five cusp form  $\Delta_5$  with the order-two Maass multiplier  $\nu_{\Delta_5}$ , and the square root, not the square, is the fundamental object. The BPS partition function  $Z_{\mathrm{BPS}}^{K_3 \times E} = \Delta_5^{-2} = \chi_{10}^{-1}$  is blind to the sign of  $\Delta_5$ ; the Pfaffian -- the square root of the determinant line -- retains the order-two orientation character, and the manuscript's central conditional identity is  $\epsilon_o = \nu_{\Delta_5}$ . Five readings of  $\Delta_5$  are assembled: the automorphic section, the denominator of a generalized Borcherds–Kac–Moody superalgebra on a rank-three Lorentzian lattice, a protected Pfaffian of a  $K_3 \times E$  factorization datum, a conjectural mirror discriminant, and the singular theta lift of the  $K_3$  half-elliptic-genus  $\phi_{o,1}$ . The triangle automorphic–denominator–theta is classical; the Pfaffian reading is a recognition criterion relative to named finite data; the mirror reading is open. This companion states the five views with exact normalisations, the two unconditional new results, the honest ledger of retained data, and the place of  $\Delta_5$  in the five-garden programme: the terminal scalar shadow of the  $K_3 \times E$  object.*

## I The seed

A partition function that is a perfect square asks for its square root. On the trivial-canonical threefold  $X = K_3 \times E$  the reduced curve-counting partition function is  $Z_{\mathrm{BPS}}^X = \chi_{10}^{-1}$ , and  $\chi_{10} = \Delta_5^2$  where  $\Delta_5$  is the unique weight-five cusp form on  $\mathrm{Sp}_4(\mathbb{Z})$  with character -- the Maass multiplier  $\nu_{\Delta_5}$ , of order two, trivial on the theta group and equal to det on the relevant reflection subgroup. Squaring kills exactly one bit: an order-two twist that no scalar function of the period sees. The manuscript's organising claim is that this bit is geometric. It is the orientation of a moduli problem -- the  $\mathbb{Z}/2$  of a Pfaffian line whose square is a determinant line -- and the five readings of  $\Delta_5$  are five levels at which one and the same  $K_3 \times E$  datum  $\mathfrak{A}_{K_3 \times E} \in E_3\text{-HolFA}(K_3 \times E)$  presents it:

- ① automorphic section      ② BKM denominator      ③ protected Pfaffian      ④ mirror discriminant      ⑤ theta lift.

The edges ①  $\leftrightarrow$  ②, ②  $\leftrightarrow$  ⑤, ①  $\leftrightarrow$  ⑤ are theorems of Igusa, Borcherds and Gritsenko–Nikulin. The edge ②  $\leftrightarrow$  ③ -- the Pfaffian–Dirac identity -- is this manuscript's contribution, and it is a *recognition criterion*: a theorem relative to a named finite package of retained data which the manuscript's own ledgers record as not yet supplied. Reading ④ is a conjecture. The five-fold thesis is deliberately not closed; what the architecture supplies is the exact list of finite data that would close it.

## 2 The objects

**Definition 2.1** (The Jacobi seed).  $\phi_{o,I}$  is the weak Jacobi form of weight  $o$  and index  $I$ , normalised by the  $K_3$  elliptic genus  $Z_{K_3}(\tau, z) = 2\phi_{o,I}(\tau, z)$ , with closed form  $\phi_{o,I} = 4 \sum_{i=2}^4 \theta_i(\tau, z)^2 / \theta_i(\tau, o)^2$  and expansion  $\phi_{o,I} = r^{-1} + 10 + r + O(q)$ . Its coefficients  $f(n, l)$  depend only on  $4n - l^2$ :

$$f(o, o) = 10, \quad f(o, \pm 1) = 1, \quad f(1, o) = 108, \quad f(1, \pm 1) = -64, \quad f(1, \pm 2) = 10, \quad f(2, o) = 808, \dots$$

with support  $4n - l^2 \geq -1$ .

**Definition 2.2** (The lattices). With  $U$  the hyperbolic plane, the ambient rank-three lattice is  $\Lambda^{2,1} = U \oplus \langle 2 \rangle$  with basis  $(f_2, f_3, f_{-2})$ ,  $(f_2, f_{-2}) = -1$ ,  $(f_3, f_3) = 2$ . The type-II sublattice  $\Lambda_{II}^{2,1}$  (even coefficients on  $f_{\pm 2}$ ) has index 4 and discriminant  $-32$ . The simple type-II roots

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3$$

have Cartan matrix  $(\delta_i, \delta_j) = 4\mathbf{I}_3 - 2\mathbf{J}_3 = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$  and Weyl vector  $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$  with  $(\rho, \delta_i) = -1$ . The off-diagonal entries  $-2$  are infinite Coxeter exponents: the reflection group  $W^{(2)}(\Lambda_{II}^{2,1}) = \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \rangle$  is the free product  $\mathbb{Z}/2 * \mathbb{Z}/2 * \mathbb{Z}/2$  -- the universal Coxeter group on three generators, with abelianisation  $(\mathbb{Z}/2)^3$ ; there is no braid relation. The theta-lift lattice is  $\Lambda^{3,2} = U \oplus U \oplus \langle 2 \rangle$ , realised as  $q_o^\perp$  inside  $\Lambda^2 \mathbb{Z}^4$  with the Pfaffian pairing; the exceptional isogeny  $\mathrm{PSp}_4(\mathbb{R}) \simeq \mathrm{SO}(3, 2)_+$  carries Siegel modular forms to orthogonal ones.

**Definition 2.3** (Mukai–Gram dictionary). The formal charge lattice of  $X = S \times E$  is  $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S, \mathbb{Z}) \otimes H^{\mathrm{even}}(E, \mathbb{Z})$ , charges  $(Q, P)$ . The quadratic Gram map  $\Pi_X(Q, P) = \left( \frac{(Q, Q)}{2}, (Q, P), \frac{(P, P)}{2} \right) \in \mathbb{Z}^3$  is not additive; its defect is a symmetric polarization  $B$  forcing the normal-ordered central extension  $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \mathbb{Z}^3$  on which the corrected map is additive (the naive additive pushforward is proved not to exist). For a curve class  $\beta_h = s_1 + hF$  with  $\chi(O_Y) = n$  and  $d$  points, the D6–D2–Do charge computes to  $\Pi_X = (h - 1, n, d - 1)$ , and the root grading is  $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$  with  $\alpha(n, l, m)^2 = -2(4nm - l^2)$ .

**Definition 2.4** (The Dirac–Igusa pro-object).  $\mathfrak{D}_X^{DI}$  is an inverse system over Harder–Narasimhan heights of fourteen-component data: the  $E_3$ -prefactorization datum, a hybrid factorization carrier on  $\mathrm{Ran}^{\mathrm{hyb}}(E) = \mathrm{Ran}(E)_{b=0} \sqcup \mathrm{Ran}^{\mathrm{wr}}(E)_{b>0}$ , the charge lattice with Gram projection, vanishing-cycle coefficients with cosection-twisted reduced  $d$ -critical orientation (the inflated  $K_3$  two-form supplies the cosection; Joyce–Upmeyer square roots supply the orientation candidates), Hall product and coproduct on  $H_c^\bullet(\mathfrak{M}^{\mathrm{red},+}(X), \Phi_W \otimes o)$ , a first-order Dirac block with Pfaffian line, a chiral Koszul source coalgebra  $C_X$  with cobar comparison towards  $U_E^{\mathrm{ch}}(\mathfrak{g}_{\Delta_5} \otimes O_E)$ , and a recognition map. Two honesty clauses are part of the definition. The “Dirac operator” is algebraic: the first-order block at degree  $\gamma$  is the  $2 \times 2$  skew matrix  $J_\gamma = \begin{pmatrix} o & 1-x^\gamma \\ -(1-x^\gamma) & o \end{pmatrix}$  with  $\mathrm{Pf}(J_\gamma) = (1 - x^\gamma)^{\mathrm{sdim} P_\gamma}$ ,  $x^\gamma = q^n r^l s^m$  -- no analytic operator on a constructed state space is asserted. And the geometric Pfaffian line satisfies  $\mathcal{L}_{\mathrm{Pf}}^{\otimes 2} \simeq \mathcal{L}_{\mathrm{det}}$ : the square root of the determinant of the cosection-reduced self-Ext complex, which is exactly where the orientation bit lives.

Three names discipline the one symbol:  $\mathcal{D}_X$  is the automorphic section (view ①);  $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1} \Delta_5(2Z)$  is the denominator (view ②);  $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$  is the protected Pfaffian (view ③, conditional). The bare handle  $\Delta_5$  never substitutes for any of the three.

### 3 The theorems

**Theorem 3.1** (View ①; proved).  $\mathcal{D}_X = \Delta_5$  is the Borcherds–Gritsenko–Nikulin section of the weight-five automorphic line on  $\mathrm{Sp}_4(\mathbb{Z}) \backslash \mathbb{H}_2$  with multiplier  $\nu_{\Delta_5}$ : weight  $5 = f(\mathfrak{o}, \mathfrak{o})/2$ , divisor the simple Humbert surface  $\mathcal{H}_1$  (the diagonal locus) with multiplicity one, leading theta coefficient  $[q^{1/2} r^{1/2} s^{1/2}] \Delta_5 = 2^6 = 64$ , and in the chamber  $\mathfrak{o} < |s| \ll |q| \ll |r|^{-1} \ll 1$  the Borcherds product

$$\Delta_5(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The multiplier has order two, restricts to  $\det$  on  $W^{(2)}(\Lambda_{II}^{2,1})$ , and is trivial on chamber-permuting type-I reflections.

**Theorem 3.2** (View ②; proved, Gritsenko–Nikulin). The generalized Borcherds–Kac–Moody Lie superalgebra  $\mathfrak{g}_{\Delta_5}$  on  $\Lambda_{II}^{2,1}$  -- three even real simple roots  $\delta_i$ , imaginary simple data with parity multiplicities read off  $\phi_{\mathfrak{o},1}$  -- has Weyl–Kac–Borcherds denominator

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \sum_{w \in W^{(2)}} \det(w) w \left( e^{-2\pi i(\rho, z)} - \sum_a m(a) e^{-2\pi i(\rho+a, z)} \right) = e^{-2\pi i(\rho, z)} \prod_{\alpha \in \mathcal{R}_+} (1 - e^{-2\pi i(\alpha, z)})^{\mathrm{smult}(\alpha)} = \frac{1}{64} \Delta_5(2Z),$$

with signed root supermultiplicities  $\mathrm{smult}(\alpha(n, l, m)) = f(nm, l)$  on the active support. The doubling  $Z \mapsto 2Z$  records the index-four sublattice; 64 is the theta normalisation, not a doubling factor.

**Theorem 3.3** (View ⑤; proved, Borcherds). The theta decomposition of  $\phi_{\mathfrak{o},1}$  is a weight  $-1/2$  vector-valued form  $h$  for the dual Weil representation of  $\langle 2 \rangle$ ; its regularized theta lift on  $\Lambda^{3,2}$ , pulled through  $\mathrm{PSP}_4(\mathbb{R}) \simeq \mathrm{SO}(3, 2)_+$ , is  $\Theta(\phi_{\mathfrak{o},1}^{\mathrm{Weil}}) = \Delta_5$ , with weight  $c_{\phi}(\mathfrak{o}, \mathfrak{o})/2$  and divisor  $\sum c_{\phi}(-\lambda^2/2, \lambda) \lambda^{\perp}$ .

**Theorem 3.4** (The BPS scalar; proved elsewhere). For primitive classes on  $K_3 \times E$  in the Oberdieck–Pandharipande chamber, the reduced Donaldson–Thomas, stable-pairs and Oberdieck–Pixton partition functions agree:

$$Z_{\mathrm{DT}}^X = Z_{\mathrm{PT}}^X = Z_{\mathrm{OP}}^X = -4096 \Delta_5^{-2}, \quad Z_{\mathrm{BPS}}^{K_3 \times E} := (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2},$$

where  $4096 = 64^2$  is the squared theta-leading constant and the sign is the Oberdieck–Pandharipande scalar branch. The arithmetic of the square:  $\Delta_{10} = \Delta_5^2$  spans the Maass space, is the Saito–Kurokawa lift of  $\Delta E_6 \in S_{18}(\mathrm{SL}_2(\mathbb{Z}))$ , with spinor  $L$ -function  $L_{\mathrm{spin}}(s, \Delta_{10}) = \zeta(s-8) \zeta(s-9) L(s, \Delta E_6)$  -- a Ramanujan-violating CAP form, eigenvalues  $\lambda(2) = 240$ ,  $\lambda(3) = 21960$ .

**Theorem 3.5** (The two unconditional new results). (i) At the triple root  $\delta_{123} = \delta_1 + \delta_2 + \delta_3$  (norm  $-6$ ) the parity-resolved multiplicities of  $\mathfrak{g}_{\Delta_5}$  are

$$\dim_{\mathfrak{o}} = 29, \quad \dim_{\bar{\mathfrak{i}}} = 93, \quad 29 - 93 = -64 = f(1, 1),$$

the 29 even directions arising as 2 independent Jacobi words  $[e_k, [e_i, e_j]]$  plus  $3 \times 9$  brackets against the nine isotropic imaginary-simple fibres (whence  $\mathrm{smult}(a_{ij}) = 10 = 1 + 9$ ), the 93 odd directions as imaginary simple generators. (ii) A divisor, a leading coefficient, a scalar Borcherds product, and its square do not determine a section of  $\mathcal{L}^5 \otimes T_{\chi}$  for a nontrivial order-two flat twist  $T_{\chi}$ : order-two monodromy is invisible after squaring. This is the precise sense in which  $\Delta_5$  carries strictly more than  $\chi_{10}$ .

**Theorem 3.6** (Pfaffian–Dirac; conditional recognition). *Granted the retained data (Do) (degeneration limit of the cosection-reduced orientation theory), (O<sub>I</sub>) (strong reduced orientation on the quotient self-Ext frame bundle), (O<sub>I</sub>)<sup>+</sup> (Weyl-equivariant transport along the three type-II reflections), (O<sub>2atlas</sub>) (the local Pfaffian wall atlas) and the finite geometric Pfaffian datum (P<sub>fin</sub>) on a cofinal Harder–Narasimhan system,*

$$\iota_{\text{aut}}(\text{Pf}_{\text{prot}}(\mathfrak{D}_X^{DI})) = \mathcal{D}_X = \Delta_5, \quad \epsilon_o = \nu_{\Delta_5} \text{ on } W^{(2)}(\Lambda_{II}^{2,1}).$$

*The orientation mechanism is a rank-one computation: at a Pfaffian wall the odd crossing complex has Pf =  $\nu u$  and the reflection flips its sign; since  $W^{(2)}$  is a free product of three involutions, a  $\{\pm 1\}$ -character is fixed by its generator values, where both  $\epsilon_o$  and  $\nu_{\Delta_5}$  take  $-1$ . Honest scope: the hypotheses of (P<sub>fin</sub>) encode the Fourier data of  $\Delta_5$ , so the identity is  $q$ -expansion rigidity relative to the datum, not a construction; the manuscript’s certificate ledgers record every component of the retained package as not supplied; and the  $2^{12} = 4096 = 64^2$  wall count is stipulated inside the atlas datum, so its match with the squared theta constant carries no independent evidence.*

**Theorem 3.7** (Primitive recognition; conditional). *Granted the ten-part finite compact-source datum (S<sub>I</sub>)–(S<sub>IO</sub>) (chain-level descent endpoint, Cartan matching, simple representatives, Borcherds–Kac relations in the Hall bracket, radical equality, no-extra-relations, generation, completed PBW, exact triangular completion, and full parity dimensions  $\dim P_{\gamma, \bar{p}} = \text{mult}_{\bar{p}}^{\text{GN}}(\alpha(\gamma))$  in both parities), the protected primitive Lie superalgebra of the  $K_3 \times E$  datum recognises the Gritsenko–Nikulin algebra,  $\mathcal{P}_X \cong \mathfrak{g}_{\Delta_5}$ , and the chiral Koszul source satisfies  $\Omega_E^{\text{ch}} C_X \simeq U_E^{\text{ch}}(\mathfrak{g}_{\Delta_5} \otimes O_E)$ .*

**Theorem 3.8** (Level-Z trace criterion; conditional). *Granted the retained trace datum (Z<sub>HB</sub>) and, as established inputs, chiral Hochschild concentration on the Koszul locus (Volume I) and the chiral-Hochschild stratification (Volume II), the bulk supertrace of the  $K_3 \times E$  chart satisfies  $\text{Tr}^{\text{bulk}}((-1)^F)_{Z_{\text{ch}}^{\text{der}}(C_X)} = \Delta_5^{-2}$ , with the Oberdieck–Pandharipande branch  $-4096 \Delta_5^{-2}$  as the separate scalar representative. The stronger level-A statement -- a gravity-line operator algebra acting on the boundary with Pentagon trace  $\Phi_{10}^{\text{un}} = \Delta_5^2$  -- is open (Volume II, second construction problem).*

**Conjecture 3.9** (View ④: mirror discriminant). *On the rank-three Mukai-invariant period component  $\Omega_{\check{M}}/\Gamma_{\check{M}} \simeq \mathbb{H}_2/\text{Sp}_4(\mathbb{Z})$  for  $M = U \oplus \langle -2 \rangle$  (Dolgachev mirror  $\check{M} = U \oplus E_7(-1) \oplus E_8(-1)$ ), the discriminant section of the mirror period family equals  $\Delta_5^{-2}$ ; its polar divisor is twice the reduced mirror discriminant, supported on the Humbert divisor where  $\Delta_5$  vanishes. The theorem-level part is the divisor identity  $[\text{div } \Delta_5] = [\mathcal{H}_1]$ ; the trivialisation, polar-order, and mirror-correspondence clauses are open.*

**Conjecture 3.10** (Twined denominators). *For  $g$  in the Mathieu group  $M_{24}$ , the  $g$ -twined BKM denominators satisfy  $\text{den}(\mathfrak{g}_{\Delta_5}^{(g)}) = \Theta(\phi_{o,1}^{(g), \text{Weil}})|_{\mathbb{H}_2}$ ; the classes  $7A, 7B, 15A, 15B$  lie outside the CHL frame (selected by  $\varphi(N) \mid 2$ ) and their constant terms need not be even -- the named arithmetic obstruction.*

## 4 The computations

The five-coordinate  $\kappa$ -tuple of  $K_3 \times E$ , five distinct constructions with no additive relation:

$$(\kappa_{\text{cat}}, \kappa_{\text{ch}}^{\text{Hodge}}, \kappa_{\text{ch}}^{\text{Heis}}, \kappa_{\text{BKM}}, \kappa_{\text{fiber}}) = (0, 0, 3, 5, 24),$$

where  $\kappa_{\text{cat}} = \chi(O_{K_3})\chi(O_E) = 0$  is Künneth-multiplicative,  $\kappa_{\text{ch}}^{\text{Hodge}} = \sum_q (-1)^q h^{0,q} = 0$ ,  $\kappa_{\text{ch}}^{\text{Heis}} = 3$  is the rank of the type-II Heisenberg–Cartan after Gram projection,  $\kappa_{\text{BKM}}(\Delta_5) = c_1(0)/2 = 5$  is the

Borcherds weight, and  $\kappa_{\text{fiber}} = 24$  is simultaneously  $\chi_{\text{top}}(K_3)$  and the Mukai rank -- numerically equal, semantically distinct. The additive identity  $\kappa_{\text{BKM}} = \kappa_{\text{ch}}^{\text{Hodge}} + \chi(O_{\text{fiber}})$  is false (it predicts 2; the value is 5): the five coordinates are incommensurable readings of one datum.

The universal Borcherds-weight identity is  $\kappa_{\text{BKM}}(\Phi_N) = c_N(o)/2$ , always naming the input denominator. On the CHL ladder  $N \in \{1, 2, 3, 4, 6\}$  the corrected weights are

$$\text{wt} = (5, 3, 2, \frac{3}{2}, 1), \quad c_N(o) = (10, 6, 4, 3, 2),$$

per Jatkar–Sen and Govindarajan–Krishna -- the weight at  $N = 4$  is half-integral, and the once-recorded ladder  $(5, 4, 3, 2, 1)$  is retracted (its error: reading the frame-shape half-sum, the weight of  $\eta(\tau)^8 \eta(2\tau)^8$ , as a Jacobi constant term). The Fake Monster form  $\Phi_{12}$  carries  $c_{\Lambda}(o) = 24$ ,  $\kappa_{\text{BKM}} = 12$ : one rule, distinct rows.

The first finite window of the denominator algebra, parity-resolved (even | odd): the isotropic doubles  $2a_{ij}$  carry  $10 | 0$ ; the triple root  $\delta_{123}$  carries  $29 | 93$ ; the next window rows carry signed dimensions  $108, -513$ , and at  $2\delta_{123}$  superdimension  $4016$  with correction  $m = -540$ . The verification  $[qrs]$ -coefficient of the monic product equals  $93 = 64 + 29$ .

Normalisation trap, stated once:  $\Phi_{10}^{\text{un}} = \Delta_5^2$  (theta-normalised) while the Oberdieck–Pixton-monic form is  $\chi_{10}^{\text{OP}} = D_5^2 = 4096^{-1} \Delta_5^2$  with  $D_5 = 64^{-1} \Delta_5$ ; the two differ by  $2^{12}$ , and every identity above names its branch.

## 5 The garden

One logarithmic 1-form generates the programme. On a smooth complex curve  $X$ , the form  $\eta = d \log(z_1 - z_2)$  is the unique coordinate-free 1-form with a residue along the diagonal of  $\text{Conf}_2(X)$ ; the Arnold relation  $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$  generates all relations among its higher companions. From  $\eta$  one builds, for an augmented chiral algebra  $\mathcal{A}$  on  $X$ , the ordered bar complex  $\overline{B}_X^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}}) \otimes \Omega_{\log}^{\bullet}(\overline{\text{FM}}_{\bullet}(X))$ ; its differential squares to zero exactly when  $\mathcal{A}$  satisfies the Borcherds identity. Assembled over  $\overline{M}_{g,n}$  with the prime-form propagator, the total differential is a Maurer–Cartan element  $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$ ; genus  $\geq 1$  flatness fails by the curvature  $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \omega_g$ , and the scalar  $\kappa \rightarrow \kappa(\text{Vir}_c) = c/2$ ,  $\kappa(V_k(\mathfrak{g})) = \dim \mathfrak{g} (k + h^{\vee})/2h^{\vee}$  -- is the central charge read as modular curvature. The Verdier–Koszul dual obeys conductor laws  $c(\mathcal{A}) + c(\mathcal{A}^!) = K$  with  $K(\text{Vir}) = 26$ ,  $K(\mathcal{W}_N) = 4N^3 - 2N - 2$ ,  $K(V_k(\mathfrak{g})) = 2 \dim \mathfrak{g}$  along  $k \mapsto -k - 2h^{\vee}$ ; the landscape stratifies into archetype classes G/L/C/M of shadow depth  $2/3/4/\infty$  plus the critical Feigin–Frenkel stratum.

Every object sits on one chain: primitive factorization category  $\rightarrow$  chart algebra  $A_b \rightarrow$  bar shadow  $\overline{B}(A_b) \rightarrow$  derived chiral centre  $Z_{\text{ch}}^{\text{der}}(A_b)$  (the bulk)  $\rightarrow$  acting line operators  $\rightarrow$  scalar modular shadow. Identifying stages without the licensing datum -- shadow = object -- is the master error the programme is disciplined against; this volume sits deliberately at the terminal stage and refuses every unlicensed promotion:  $\Delta_5$  is not a Hilbert space,  $Z_{\text{BPS}}$  is not a path integral, the denominator is not the bulk.

The five gardens are five readings of the chain. *Volume I* (chiral-bar-cobar) builds the engine on algebraic curves: bar-cobar adjunction and inversion, quadratic Koszul recognition, centre complementarity, the deformation–Hodge–graph traces  $F_g = \kappa \lambda_g^{\text{FP}} + \partial F_g^{\text{cross}}$ , Hochschild concentration, and the arithmetic shadow. *Volume II* (chiral-bar-cobar-vol2) reads the same object as physics: bar = BV/BRST, quantum master equation = Maurer–Cartan, anomaly = curvature ( $c - 26 \leftrightarrow \kappa_{\text{tot}}$ ), the two-coloured Swiss-cheese operad as bulk–boundary home, and Universal

Holography  $\mathcal{A} \mapsto (\partial = \mathcal{A}, \text{bulk} = Z_{\text{ch}}^{\text{der}}(\mathcal{A}))$ , at  $\text{Vir}_{c=3\ell/2G_N}$  the boundary reading of 3d gravity. *Volume III* (calabi-yau-quantum-groups) constructs the geometric source: the two-stage functor  $\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}}$  from Calabi–Yau  $d$ -categories to chiral algebras on a curve, the cohomological Hall algebra  $Y^+(X)$  with Drinfeld double  $G(X)$  as the level-three quantum group -- at  $K_3 \times E$  the conditional Hall–Borcherds recognition of  $\mathfrak{g}_{\Delta_5}$ , that this volume’s criteria make precise. *Mixed holomorphic-topological strings* computes the local germ of the  $d = 2$  theory at  $N$  Dirac branes,  $J(f) = \text{Tr } f(\phi_1, \phi_2)$ , with the holomorphic de Rham obstruction  $H^1(K_3 \times E, \mathcal{O}) \neq 0$  showing exactly why the local-to-global route needs more than the formal stalk. *This volume* is the terminal scalar: the pentadic  $\Delta_5$ , the one order-two bit that survives every projection, and the finite list of data that would promote it back to an operator.

The single question underneath: characterise the modular convolution algebra  $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$  as a geometric object. The proposed answer -- endomorphisms of the Lagrangian embedding  $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\text{vac}}(\mathcal{A})$  in a  $(-2)$ -shifted symplectic stack of vacua -- has Volume I computing its Taylor jets, Volume II its local composition law, Volume III its geometric source, and this volume its sharpest example: the  $K_3 \times E$  datum whose scalar shadow is  $\Delta_5$ .

## 6 The frontier

The manuscript’s architectural invention is the recognition-criterion format: every conditional theorem names the exact finite data that would close it, and the ledgers record what is supplied. The open list, in order of depth:

- (i) The finite compact-source datum  $(S_I) - (S_{IO})$  for primitive recognition, and the finite Koszul comparison datum -- the operator-level construction of  $\mathfrak{g}_{\Delta_5}$  from  $K_3 \times E$  moduli.
- (ii) The orientation package:  $(D_{\text{OHN}})$  degeneration datum,  $(O_{\text{Iquot}})$  quotient orientation, Weyl transport  $\mathcal{T}_W$ , and the type-II wall atlas  $(O_{2\text{atlas}})$  -- what makes  $\epsilon_0$  a character globally.
- (iii) The finite geometric Pfaffian datum  $(P_{\text{fin}})$ , replacing  $q$ -expansion rigidity by construction.
- (iv) The trace datum  $(Z_{\text{HB}})$  and the Hall–Borcherds residual -- the four-part chain datum whose derived-centre trace is  $\Delta_5^{-2}$ .
- (v) The gravity-line operator algebra with Pentagon trace  $\Delta_5^2$  (Volume II’s second construction problem) -- the level-A completion.
- (vi) The mirror clauses  $(M_1) - (M_4)$  of view ④.
- (vii) Twined denominators for all twenty-six  $\mathcal{M}_{24}$  classes, with the parity obstruction at the non-CHL classes.

What stands unconditionally: the classical triangle ①②⑤ with every normalisation pinned; the Oberdieck–Pandharipande scalar  $-4096 \Delta_5^{-2}$ ; the Saito–Kurokawa arithmetic of the square; the parity-resolved window  $29 \mid 93$  with  $29 - 93 = f(1, 1)$ ; and the monodromy proposition that makes the square root a genuine invariant. The pentad is a programme; the square root is already a theorem.